

Developing a Low-Cost Wireless Transtibial Prosthetic Sensor Sock

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Current Bottleneck in the Prosthetic Fitting Process

Fitting a below the knee prosthetic socket is a painful process that spans multiple appointments. Clinicians work with subjective patient feedback, and visual observation, with no objective data. For the patient, this process means weeks of discomfort and pain during a very sensitive time in their recovery. This correlates with an increase in patient dissatisfaction which can lead to prosthetic abandonment.

Tools to quantify prosthetic fit exist, but not in a form clinics can use. Commercial sensor arrays have a cost barrier, and with accuracy errors up to 33% on the curved geometry of a residual limb. Most solutions tether the patient to a computer, which rules out dynamic analysis, with software that returns raw numbers rather than a clinician friendly visual.

Sensor Architecture

A Force Sensitive Resistor decreases in resistance as force is applied. They are .1 mm thick which is ideal for comfort. These sensors can be paired with a resistor to read the output voltage, which correlates to an applied pressure after calibration.

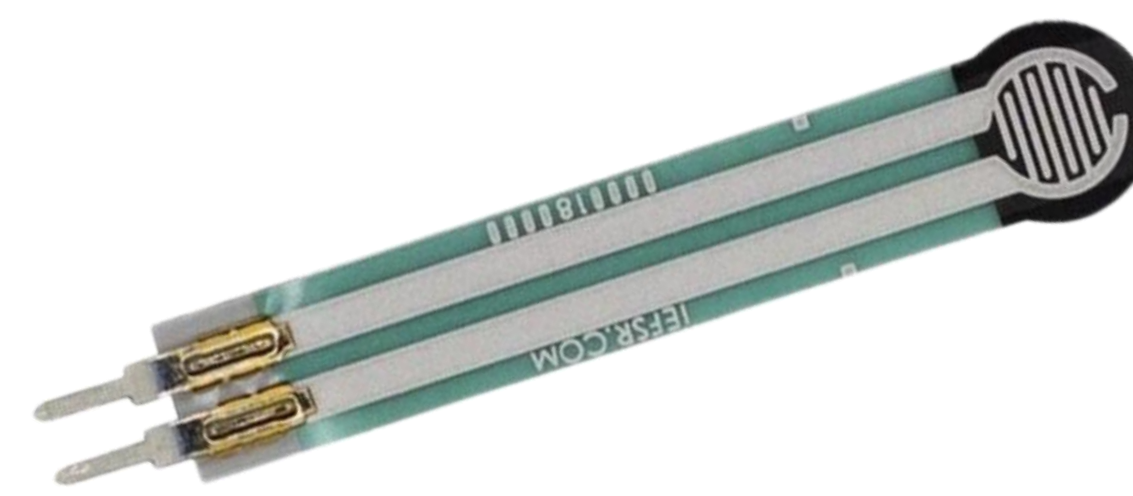


Figure 1. An FSR 400 with the active sensor area enclosed in black

Our Solution: Wireless, Affordable, & Comfortable

The current platform has integrated the sensors, hardware, and data visualization into a single package for bench testing. We will improve on this design and begin clinical trials.



Figure 2. 3D-printed limb stand with the installed sensor array

Current Prototype: We have developed an array of 16 Force Sensitive Resistors (FSRs) stitched onto a prosthetic sock. These sensors convert applied pressure into a voltage. This sensor array connects to a pair of custom PCBs that feature an ESP32 for data acquisition and transmission. With battery power and a Wi-Fi connection, this tether-free solution resolves many mechanical issues faced by commercial solutions.

For our next iteration, our biggest focus will be a smaller form factor. Both PCBs will be combined into one, small enclosure for greater comfort. We will also upgrade the sensors for a wider sensing range and greater accuracy.

Software: The program features a 3D heatmap to display real-time pressure distribution across the limb. Live values for each sensor are displayed in the bar graph. Load cycling over time can be viewed in the line graph. User customization is available with recordings and replays of clinical trials.

These features were set to match clinician requests and are currently available in a MATLAB simulation with plans to develop a commercial software.

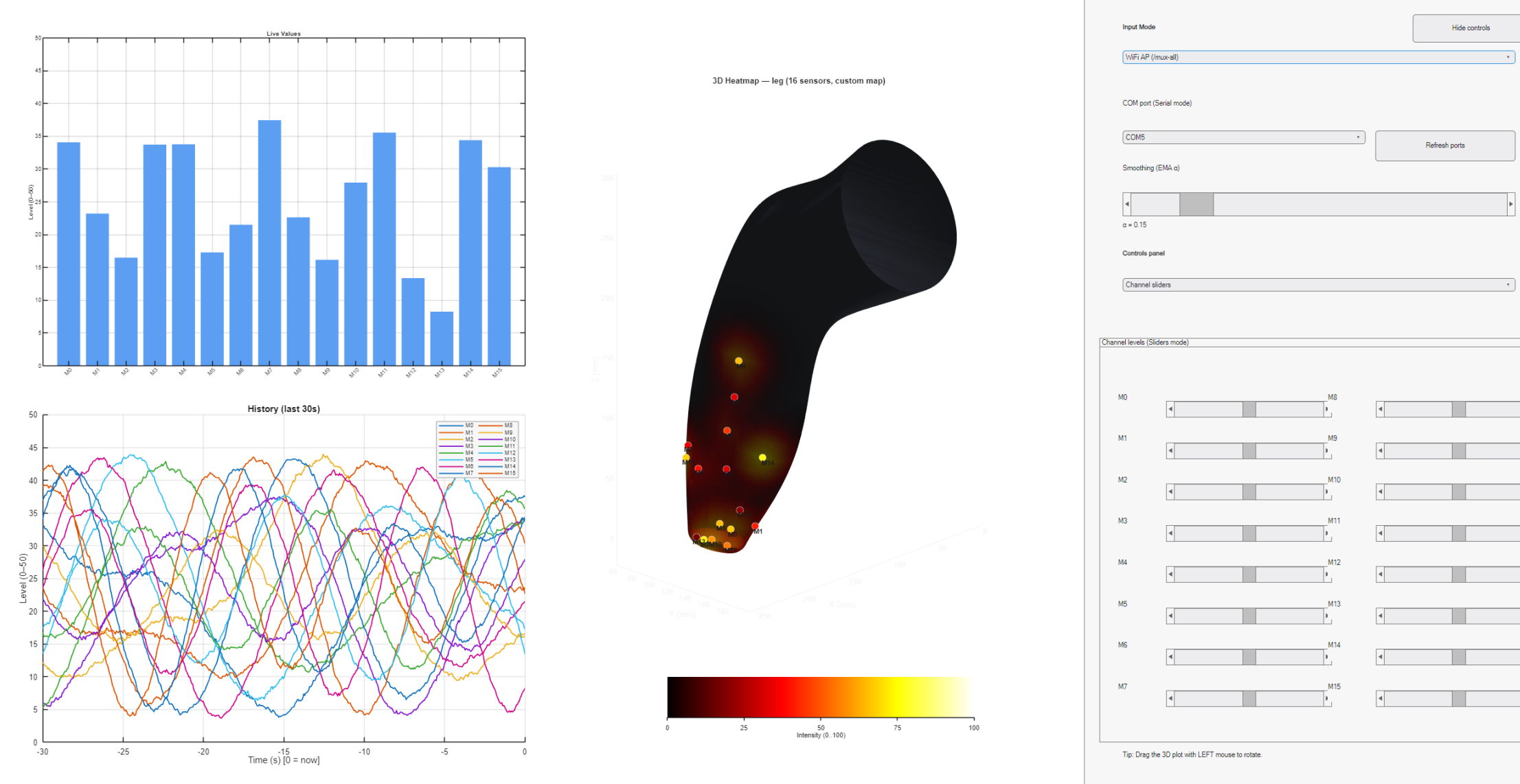


Figure 3. MATLAB GUI with graphs, visualizations, and user controls

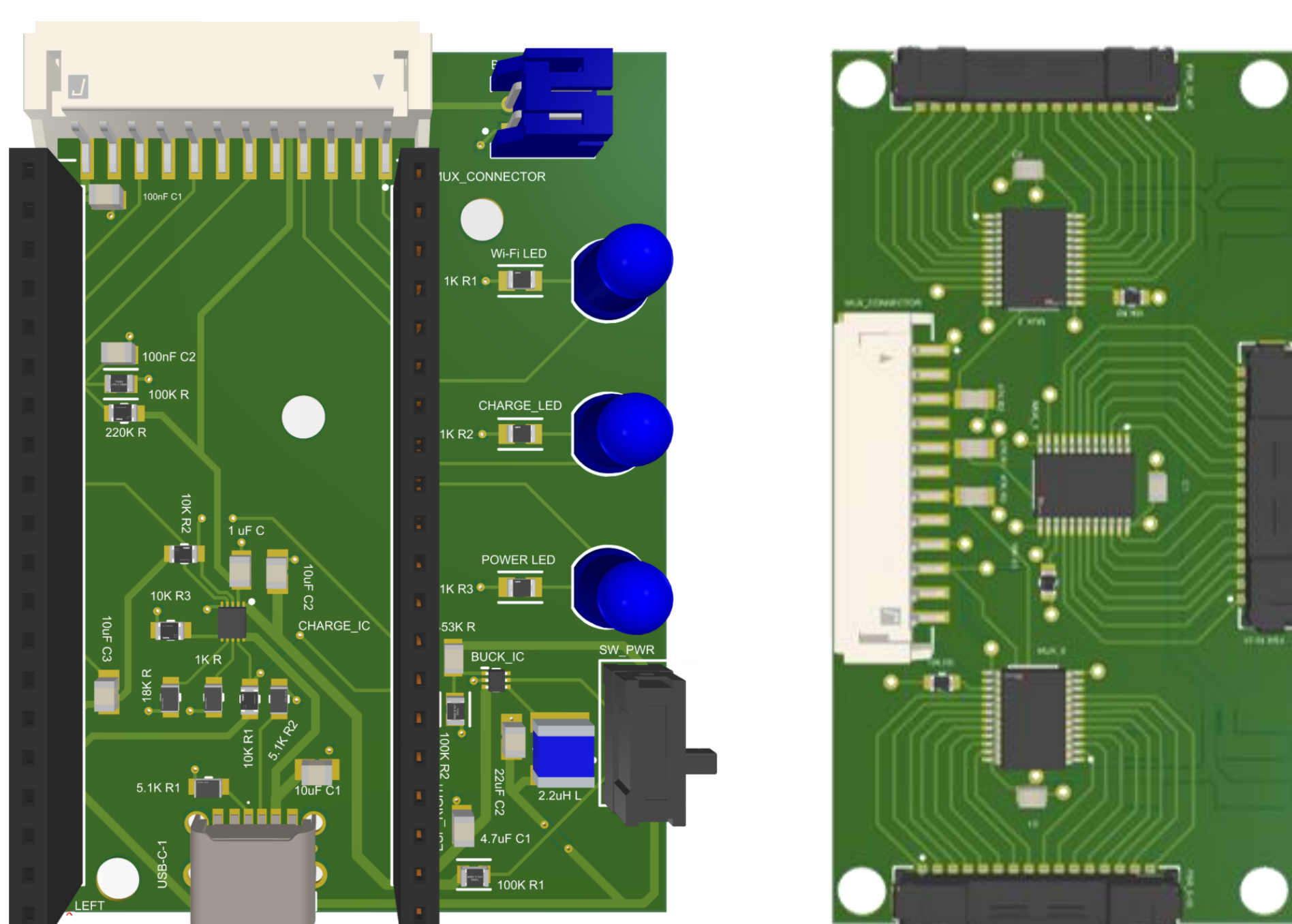


Figure 4. 3D render of the ESP32 and analog multiplexer PCBs

Hardware: A custom PCB was designed featuring an ESP32 to collect and transmit sensor readings. The left PCB hosts the ESP32, user LEDs, and the USB-C chargeable battery, which is strapped to the thigh. This PCB connects to the clinician's computer by connecting to the local Wi-Fi network.

The second PCB on the prosthetic sock features a three 16:1 multiplexers capable of reading up to 48 FSRs simultaneously.

Verification and Validation

Our clinical collaboration ensures that our solution can be seamlessly integrated with patient care.

- Various sensor locations have been tested to focus on sensitive locations and provide equal coverage.
- Our FSRs will be pressure calibrated to ensure reliable, consistent performance over their operating lifetime.
- The ESP32 enable wireless data transmission, which solves the issue of tether management from previous commercial solutions.
- Our software focused on providing detailed metrics that are easier to visualize.

Future Goals for Commercialization

Before we head into clinical testing, there are a few areas of improvement to focus on:

- The two PCBs will be combined into one for a smaller form factor while improving comfort by moving all components above the knee.
- A smaller enclosure will be strapped to the knee to provide a more realistic testing environment for the patient.
- The 16 FSRs will be upgraded to a newer sensor generation with a larger pressure sensing range and higher definition.
- We are considering a flexi-PCB solution to replace wires for a more premium product.

References & Acknowledgments

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